

Problem Statement

Educational institutions need an efficient way to monitor and evaluate student academic performance. Currently, student information such as marks, percentage, rank, and subject-wise performance is scattered across multiple records, making it difficult for teachers and administrators to quickly identify top performers and students needing improvement.

The lack of a centralized and interactive system limits effective performance tracking, subject analysis, and timely academic support. Teachers also face challenges in analyzing class-wise and section-wise performance trends and providing personalized feedback to students.

The Student Performance Dashboard is designed to provide a clear and interactive view of student academic data, enabling educators to track marks, rankings, percentages, and weak subjects efficiently for better decision-making and student improvement.

Dashboard Requirements

1. Student Information Requirements

The dashboard should display:

- Student ID
 - Student Full Name
 - Class
 - Section
 - Student Photo/Profile Icon
-

2. Academic Performance Requirements

KPI Cards

The dashboard should show:

- Total Marks
 - Rank
 - Percentage
 - Academic Status (Distinction/Pass/Fail)
-

3. Subject-wise Performance Analysis

The dashboard should display marks scored in subjects such as:

- Hindi
- English

- Maths
- Social
- Science
- Telugu

Requirements:

- Compare marks across subjects
 - Highlight weak-performing subjects
 - Identify strongest subjects
-

4. Performance Visualization Requirements

The dashboard should include:

- Bar Chart for subject-wise marks
 - Gauge Chart for overall percentage
 - Summary/Insight section showing:
 - Student achievement
 - Improvement areas
-

5. Filter Requirements

The dashboard should allow filtering by:

- Student Name
 - Class
 - Section
-

6. Functional Requirements

The dashboard should:

- Provide interactive student performance analysis
 - Help teachers identify students needing improvement
 - Support quick academic evaluation
 - Enable easy comparison of subject performance
-

7. Business/Academic Requirements

The dashboard should help:

- Teachers monitor student progress
- Schools track academic performance
- Identify weak subjects early
- Improve student learning outcomes
- Support data-driven academic decisions